OSHA STANDARD 1910 REGULATORY SAFETY MANUAL

United States Occupational Safety and Health Administration (OSHA)
29 CFR Part 1910 - General Industry EHS Regulatory Compliance Corpus

---

1. OSHA 1910.132 - General Requirements for PPE

Hazard Assessment & Mandatory Protective Equipment

Protective equipment for eyes, face, head, and extremities, protective clothing, and protective shields shall be provided, used, and maintained in a sanitary condition. Employers must verify workplace hazard assessment with written certification.

- Halt personnel entering active industrial area without verified PPE.
- Verify ANSI Z89.1 head protection and ANSI/ISEA 107 high-visibility vest.
- Inspect safety equipment for structural degradation before re-authorizing access.

EMERGENCY PROTOCOL (SOP-EHS-042): Halt line operations and equip standard hard hat and high-vis apparel immediately.

2. OSHA 1910.135 - Head Protection (Safety Helmets & Hard Hats)

Protective Helmets in Overhead & Falling Object Zones

The employer shall ensure that each affected employee wears a protective helmet when working in areas where there is a potential for injury to the head from falling objects or electrical shock hazards. Helmets must comply with ANSI Z89.1 Type I/II, Class E/G standards.

- Mandatory hard hats in crane hoist zones, warehouse pallet racks, and construction aisles.
- Inspect shell for hairline fractures and suspension harness clearance.
- Immediate ejection of unhelmeted workers from active overhead swing radius.

EMERGENCY PROTOCOL (SOP-EHS-042): Trigger localized safety alarm and enforce immediate helmet compliance.

3. OSHA 1910.132(d) & ANSI/ISEA 107 - High-Visibility Reflective Apparel

Reflective Garments in Heavy Equipment & Vehicle Transit Zones

Personnel exposed to industrial trucks, forklifts, or heavy machinery shall wear Class 2 or Class 3 high-visibility safety vests with 360-degree retroreflective striping visible up to 1,000 feet.

- Require high-visibility vest in loading bays, transit pathways, and staging docks.
- Direct non-vested workers to marked green pedestrian aisles.
- Ensure fluorescent background fabric complies with chromaticity standards.

EMERGENCY PROTOCOL (SOP-EHS-045): Redirect personnel to safe pedestrian lane and equip Class 2 safety vest.

4. OSHA 1910.178 - Powered Industrial Trucks (Forklifts)

Pedestrian Clearance Buffer & Operational Safety Rules

A minimum buffer of 3.0 meters (10 feet) must be maintained between moving forklifts and pedestrian workers. Forklift operators shall sound directional horns at intersections and adhere to facility speed limits (<5 mph).

- Strobe activation upon pedestrian entry into 3-meter turning radius.

- Operator must lower forks to floor, set emergency brake, and cut engine.
- Pedestrian must utilize physical bollard-protected crosswalks.

EMERGENCY PROTOCOL (SOP-EHS-077): Audible horn trigger & full stop of forklift until pedestrian clears zone.

5. OSHA 1910.179 - Overhead and Gantry Cranes

Suspended Load Fall Zone Protocols & Hoisting Directives

No employee shall be permitted beneath an elevated hoist or suspended load. Operators must sound an audible alarm before trolley traverse and enforce a 10-meter exclusion zone.

- Automated laser boundary monitoring beneath active hoist hooks.
- Immediate trip of hoist wireless E-STOP upon perimeter breach.
- Rigging inspection by qualified safety specialist prior to restart.

EMERGENCY PROTOCOL (SOP-EHS-091): EMERGENCY HOIST E-STOP. Evacuate 10m perimeter around suspended load.

6. OSHA 1910.147 - Control of Hazardous Energy (Lockout/Tagout)

Machine Isolation & Zero Energy State Protocol

Unauthorized access into automated machinery cells or energized electrical panels is strictly prohibited. Procedures must ensure zero energy state using standardized padlocks and tags.

- Execute instantaneous E-STOP upon optical light curtain breach.
- Verify zero residual pneumatic, hydraulic, and electrical charge.
- Authorized technician must apply Lockout/Tagout padlocks before servicing.

EMERGENCY PROTOCOL (SOP-EHS-108): DE-ENERGIZE SECTOR. Cordon perimeter and verify zero energy state.

7. OSHA 1910.28 / 1910.29 - Walking-Working Surfaces & Fall Arrest

Leading Edges, Elevated Platforms & Guardrail Requirements

Unprotected sides or edges 4 feet (1.2m) or higher require compliant guardrail systems, safety nets, or personal fall arrest systems with 5,000 lb rated anchor points.

- 100% tie-off mandatory on scaffoldings and elevated catwalks.
- Toe-boards minimum 3.5 inches to eliminate falling tool hazards.
- Inspect harnesses and self-retracting lifelines prior to daily shift.

EMERGENCY PROTOCOL (SOP-EHS-058): Secure worker with certified fall arrest lifeline and inspect guardrail gates.